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Dynamic exposure and body burden models for per- and polyfluoroalkyl substances (PFAS) enable management of food safety risks in cattle

Antti T. Mikkonen^{a,b,*}, Jennifer Martin^b, Richard N. Upton^a, Jan-Louis Moenning^c, Jorge Numata^c, Mark P. Taylor^b, Michael S. Roberts^{a,d}, Lorraine Mackenzie^{a,e}

^a University of South Australia, Clinical and Health Sciences, Adelaide, South Australia, Australia

^b Environment Protection Authority Victoria, EPA Science, Centre for Applied Sciences, Macleod, Victoria 3085 Australia

^c German Federal Institute for Risk Assessment BfR, Department Safety in the Food Chain, Max-Dohrn-Str. 8-10, 10589 Berlin, Germany

^d University of Queensland, Frazer Institute, Queensland, Australia

^e Basil Hetzel Institute for Translational Health Research, The Queen Elizabeth Hospital, Woodville South, South Australia, Australia



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ABSTRACT

With increasing global focus on planetary boundaries, food safety and quality, the presence of per- and polyfluoroalkyl substances (PFAS) in the food chain presents a challenge for the sustainable production and supply of quality assured food. Consumption of food is the primary PFAS exposure route for the general population. At contaminated sites, PFAS have been reported in a range of agricultural commodities including cattle. Consumer exposure assessments are complicated by the lack of validated modelling approaches to estimate PFAS bioaccumulation in cattle. Previous studies have shown that PFAS bioaccumulation in livestock is influenced by environmental, spatial and temporal factors that necessitate a dynamic modelling approach.

This work presents an integrated exposure and population toxicokinetic (PopTK) model for cattle that estimates serum and tissue concentrations of PFAS over time. Daily exposures were estimated from intakes of water, pasture, and soil, and considered animal growth, seasonal variability (pasture moisture content and temperature) and variable PFAS concentrations across paddocks. Modelled serum and tissue estimates were validated against monitoring data from Australian and Swedish cattle farms. The models were also used to develop and test practical management options for reducing PFAS exposure and to prioritise remediation for farms. Model outputs for exposure management scenarios (testing cattle rotation and targeted supplementation of feed and water) showed potential for marked reductions in consumer exposures from cattle produce.

1. Introduction

The last decade has seen a marked growth in our understanding of the environmental distribution and fate of per- and polyfluoroalkyl substances (PFAS) (Guelfo et al. 2021; Adamson et al. 2022; Evich et al. 2022; Lee and Mabury). However, the tools to estimate and mitigate risks from terrestrial food chain exposures are still extremely limited. PFAS are a class of artificial compounds, which have garnered an increasing level of attention from members of the public and regulators alike due to their pervasive presence in the built- and natural environments and their individual capacities to bioaccumulate and affect the health of consumers. Some legacy PFAS, including perfluorooctanesulfonic acid (PFOS), are ubiquitously distributed and have been detected in wildlife and human blood sera across the globe (Giesy and Kannan 2001; Kannan

et al. 2004; CDC 2019; Toms et al. 2019; EFSA 2020; Fenton et al. 2020 Dec 7; Robuck et al. 2020; ATSDR 2021). In light of this, agencies across the globe have adopted regulatory limits in an attempt to manage PFAS exposure from the environment (water and soil) and recently some jurisdictions have also developed maximum levels, trigger points or action levels (depending on the agency) for PFAS concentrations in food (FSANZ 2017; Smith and Simones 2020a; EC 2022). For the general population, diet has been consistently shown to be the major contributor of PFAS exposure, estimated to account for 66%–100% of PFOS (Lorber and Egeghy 2011; Gebbink et al. 2015; EFSA 2018; Sunderland et al. 2019). Meat-, dairy- and egg products were listed as important contributors to chronic human exposure in the European Food Safety Authority's (EFSA) dietary exposure assessment (with over 90,000 analytical results for food samples) (EFSA 2020). Animal products are

* Corresponding author: University of South Australia, Clinical and Health Sciences, Adelaide, South Australia, Australia.

E-mail address: antti.mikkonen@epa.vic.gov.au (A.T. Mikkonen).

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implicated as the key contributors due to their higher potential to bioaccumulate some PFAS (like PFOS) from the environment. The bioaccumulation potential of a particular PFAS is related to how quickly it is absorbed relative to its elimination, with bioaccumulation potential typically increasing with chain length in animals (Ng and Hungerbühler 2014).

PFAS contamination in agricultural areas is not uncommon, due to a number of factors like soil amendment with contaminated biosolids, use of contaminated water for irrigation or proximity to source sites (sites that use, produce or handle PFAS products, byproducts or waste) (Costello and Lee 2020 Dec 15; Jha et al. 2021). In many parts of the world, land application of biosolids in agriculture is a common practice for sustainable nutrient resource management and food production (Rosemarin et al. 2020; Ippolito and Barbarick 2022). There is also a growing need to increase the use of recycled water to support agriculture and increase climate change resilience in food security, resulting in policy and government sponsored programs to increase its use (AgVIC 2018). Subsequent use of PFAS contaminated paddocks and water resources for livestock grazing and feed production has resulted in the transfer of PFAS to livestock (Kowalczyk et al. 2013; Vestergren et al. 2013; Bräunig et al. 2017; Death et al. 2021). Although total diet surveys have shown that PFAS are typically low in individual foods (EFSA 2020; FDA 2021; FSANZ 2021), the EFSA has concluded based on occurrence studies and serum levels that parts of the European population exceed the tolerable weekly intake (TWI). Exposure may be even higher in localized settings within farming communities where consumption of locally sourced animal products could be higher. Understanding local exposure and whether it can be mitigated poses a challenge for environmental authorities to enact proportionate regulatory intervention at contaminated sites. Regulators face similar problems around the world and in some cases, the discovery of PFAS contamination on agricultural land and in livestock, has led to extreme measures like the termination of contaminated herds and loss of livelihoods (The Guardian – “This has poisoned everything”, February 2019 (Linn 2019 Feb 20)). One of the complicating factors for risk assessment is that it is difficult to predict site-specific PFAS exposures and resulting body burdens for cattle. Based on the experience of EPA Victoria, as a member of an interagency group dealing with PFAS impacted farms, modelled serum PFAS results rarely coincide with measured serum concentrations for cattle, which reduces reliance on modelling approaches for decision-making. Existing modelling approaches have typically assumed that PFAS exposures are constant (linked to daily intakes for feed-, water- and soil) and assume steady state kinetics (Smith and Simones 2020b; Johnston). However, we have recently reported that in addition to environmental factors, PFAS exposure may be influenced by seasonal- and behavioral patterns necessitating a dynamic modelling approach (Mikkonen et al. 2023 Feb). Fixed (or bodyweight scaled) exposure assumptions cannot capture the effect of pathways with seasonal variability, like drinking water. Dynamic exposure estimates in turn require dynamic biological models to account for ingested and excreted PFAS and the change in body burden over time. These types of models, also known as pharmacokinetic-, toxicokinetic-, and physiologically based pharmacokinetic (PBPK) models have been developed for several species (including rats, monkeys, dairy cows, pigs, chickens (Loccisano et al. 2012; Van Asselt et al. 2013; Numata et al. 2014; Worley and Fisher 2015; Chou and Lin 2019; Kowalczyk et al. 2020)) as well as humans (Loccisano et al. 2013; Fàbrega et al. 2014; Goeden et al. 2019). The primary focus of these models has been on simulation of body burdens using dichotomous or simplified exposure assumptions. Exposure to environmental pollutants, especially in extensive agriculture, may be highly variable (Ashauer and Brown 2013) and this presents an important challenge to develop and integrate dynamic exposure models for these types of pollutants (like PFAS) with biological models. Previous studies have considered dynamic PFAS human exposure from drinking water using a one-compartment toxicokinetic model (Goeden et al. 2019; Chiu et al. 2022). However, similar works have not been published for food

production animals. In order to develop site-specific mitigation or management options for cattle rearing, modelling methods must be able to account for how the different exposure variables contribute to PFAS body burden throughout the life cycle of the animals. For example, to accommodate cattle rotation plans aimed to reduce PFAS body burden, exposure models must reflect the environmental concentrations and resulting exposures from different paddocks.

Thus, in response to the identified gap in integrated dynamic exposure and biological modelling tools for cattle, this study sought to develop an integrated exposure and population toxicokinetic (PopTK) modelling approach to estimate site-specific PFAS exposure and bioaccumulation in beef cattle over time. This modelling approach is intended to be used in conjunction with regulatory limits to determine if the PFAS levels in beef produce from a site under investigation are acceptable. If PFAS levels are not acceptable, site-specific mitigation measures can be modelled to determine if acceptable levels are achievable or if operations should be ceased until remedial actions can be undertaken. This study focuses on PFOS and PFHxS, as they typically account for a high proportion of the total measurable PFAS body burden of cattle, making them the primary risk to food quality and safety of consumers in affected areas (Drew et al. 2021 Nov 3; Lupton et al. 2022; Mikkonen et al. 2023 Feb). Although PFOA is a contaminant of concern and is often detected in the environment, it is rarely detected in beef cattle (Mikkonen et al. 2023 Feb) due to its short half-life (Lupton et al. 2011); likewise PFOA displays comparatively much lower transfer rates into cow's milk (Kowalczyk et al. 2013). For these reasons, PFOA is not included in this modelling work. Studies related to PFAS bioaccumulation using integrated dynamic exposure and PopTK-modelling have so far, to the best of our knowledge, not been published for food production animals.

2. Methods

2.1. Exposure dynamics

Grazing livestock are exposed to environmental contaminants such as PFOS and PFHxS via ingestion of contaminated water, feed, and soil. Exposure estimation for grazing cattle is intrinsically linked to dry matter intake (DMI) (NRC 2000; Lukas et al. 2008). DMI is an estimate of herbage intake which indirectly influences water intake, as well as soil ingestion during grazing (NRC 2000; Lukas et al. 2008). DMI in turn is influenced by seasonal variability, pasture composition and herbage availability, which can vary considerably for pasture-based systems, with dry matter content ranging from 15% in winter up to 90% in late summer in Victoria, Australia (Smith et al., 1998a,b; Cullen et al. 2008; Garcia et al. 2008). Dry matter intake (pasture) was estimated using Eq. (1) developed by Freer et al. (Freer et al. 2012) based on relative size and condition of animals.

$$DMI_{Max}(kg/day) = C_{rs} \times MW \times Z \times (C_q - Z) \times \left(C_{rc} \times \frac{C_{bc} - C_{rc}}{C_{bc} - 1} \right) \quad (1)$$

where C_{rs} is a relative size scalar constant (0.025 kg/kg for cattle), MW is mature weight (MW may vary depending on breed; if unknown 640 kg may be used as a pooled average (Jenkins et al. 1993)), Z is the ratio of normal weight to MW (estimated based on current body weight), C_q is a quadratic constant for relative size (1.7; unitless), C_{rc} is a constant that describes relative condition (1); unitless) and C_{bc} is a constant that describes the effect of body condition on DMI (1.5; unitless).

Considering cattle go to market at different life stages, it is important to estimate exposure over the relevant timeframes. Growth (BW) can be estimated using the regression relationship Eq. (2) first described by Brody (Brody 1945) and allometric scaling constants described by Taylor (Taylor 1968).

$$BW(kg) = MW - (MW \times 0.9) \exp(-C_{gr} t / MW^{C_{gr}}) \quad (2)$$

where MW is mature weight, $C_{gr}=0.0115 \text{ kg}^{0.27} \text{ d}^{-1}$ is the growth rate constant, t is time (days), and $C_{grs}=0.27$ (dimensionless) is the allometric scalar for growth rate.

Total daily water intake (DWI) is described by Eq. (3) as the sum of free water intake (or FWI from drinking), pasture water intake (PWI from grazing) and water produced from metabolic (met) processes (such as digestion of high energy grains), which accounts for a maximum of 5% total daily water intake (NRC 2000).

$$DWI(L/day) = FWI + PWI + met \quad (3)$$

PWI can be estimated with Eq. (4) using DMI and information about pasture conditions (dry matter content).

$$PWI(L/day) = (DMI/DM_{frac}) \times (1 - DM_{frac}) \quad (4)$$

where DM_{frac} is the dry matter content of the pasture (as a fraction). The dry matter content of pasture is pasture species, site (region) and season specific. DM_{frac} can be analysed from point-in-time measurement of specific farm pastures or estimated from regional monthly data for pasture dry matter content. DM_{frac} used in this work is based on pasture conditions typically experienced in SW Victoria with data provided in the [Supplementary Materials \(Table S2\)](#).

FWI is the amount of water required in addition to PWI to meet total daily water requirements. DWI is influenced by a number of variables including type of feed, BW, activity and temperature (NRC 2000). The primary external driver for DWI, in temperate and semi-arid climate zones (like that experienced in much of the grazing regions of Victoria, Australia) is temperature, which has an effect on pasture moisture as well as increasing the physiological need for total water intake. No published regression relationships were identified for cattle which account for variable water needs associated with temperature. As such, total daily water intakes reported in NRC (2000) for beef cattle (by BW, sex and temperature) were used in a regression analysis. Multiple regression analysis was undertaken using Cubist (Kuhn and Quinlan 2020) in R (computing software). FWI was estimated by subtracting PWI from DWI.

In addition to PFAS exposure via drinking water, livestock may be exposed to PFAS via forage intake (from accumulation in the plant biomass) and incidental ingestion of soil. Depending on the site and agricultural practices, the driver(s) for exposure may vary; however, as reported by Mikkonen et al. (2023) contaminated drinking water sources tend to lead to higher body burdens in grazing cattle in Victoria, Australia. This may not be the case for every site; in some instances, application of PFOS containing industrial waste (to crop pastures) has led to elevated cattle body burdens (up to 2000 ng/ml) (Kowalczyk et al. 2013; Beecher 2019). Exposure from forage ingestion (DMI) was estimated using measured pasture concentrations as well as site specific soil-to-plant transfer factors (TFs) to estimate plant concentrations. Soil intake (SI) was also estimated for incidentally ingested soil during grazing. Herlin and Anderson (1996) collated data from published studies on soil ingestion rates in cattle under different conditions and reported that soil ingestion rates ranged from 1.2 to 17.9% of DMI and were highly variable depending on season, level of supplementary feeding and type of cattle (dairy versus beef). Soil intake can be more accurately estimated with information on herbage digestibility and monitoring data for titanium dioxide in soil and faeces (Herlin and Andersson 1996). In the absence of titanium dioxide measurements, a conservative estimate of 10% DMI was used to account for soil intake in this work.

The daily dose (of PFOS and PFHxS) from FWI (water), DMI (pasture) and SI (soil) was then estimated using environmental monitoring data for the given site. The sequence of calculations to account for exposure dynamics were coded as an R script to generate a lookup table.

2.2. Toxicokinetic model

Toxicokinetics describes the time course for absorption, distribution, metabolism and excretion (commonly referred to as ADME) of toxicants/xenobiotics. Toxicokinetic models are used to predict the concentration–time profiles for toxicants (Aarons 2005). Model design should reflect the simplest explanation that fits the evidence and suits the needs of the end-user. In this case, a one-compartment (1st-order) toxicokinetic model was adopted for its simplicity and relative ease use. Similar models have been used previously to describe the relationship between dynamic dose and serum levels (Goeden et al. 2019). A two-compartment model of seven PFAS in growing pigs was also demonstrated to be reducible to a one-compartment model (Numata et al. 2014). This model assumes that once ingested, PFAS is rapidly absorbed and equally distributed within the body (with the body being represented by a central compartment with a volume of distribution). The amount in the central compartment (AC) at a given time is a function of the dose (over that period) and the elimination rate, and can be estimated using the differential equation

$$d(AC(t))/dt = D_{daily} - kE \times AC(t) \quad (5)$$

Here, AC is the amount (μg) in the central compartment at a given time, defined as the total amount of available PFAS in the animal's body. D_{daily} is the ingested daily dose (μg) and kE is the first order elimination rate (1/d) estimated based on the half-life (DT_{50}) as detailed below. Eq. (5) includes the assumption that the absorption is 100%, based on oral absorption studies with rats (Chang et al. 2012) and mass balance estimates in a cattle feeding trial (Kowalczyk et al. 2013).

Serum PFAS concentrations are given by $C_{serum} = AC/V_d$, where V_d is the (apparent) volume of distribution (L). V_d represents the theoretical volume necessary to contain the amount AC given the observed concentration in blood serum. No prior studies reporting the V_d in livestock were identified, so the volume of distribution normed by body weight $\hat{V}_d$ (L/kg, defined as Eq. (6)) was fitted using the FME (Soetaert and Petzoldt 2010) package to published serum data from feeding studies with PFOS and PFHxS in cattle (Kowalczyk et al. 2013) (See [Supplementary Section Volume of Distribution](#)). V_d (volume of distribution, L) and kE were adjusted per Eqs. (6) and (7) for growth (body-weight (BW) and mature weight (MW)) using allometric scaling (scaling coefficient 1.0 applied to physiological volumes and 0.25 to time-related indices) (Knibbe et al. 2005) and are now not constants but functions of time.

$$V_d(t)(L) = \hat{V}_d \times BW(t) \quad (6)$$

$$kE(t)(1/d) = \ln(2) / \left[DT_{50} \times ((BW(t))/MW)^{0.25} \right] \quad (7)$$

This model was pragmatically parametrised using serum concentration, so that Eq. (5) becomes

$$d(C_{serum}(t)V_d(t))/dt = D_{daily} - kE(t) \times C_{serum}(t)V_d(t) \quad (8)$$

Although PFAS are known to partition differently between serum, plasma and whole blood (e.g. in humans (Jian et al. 2018)), the current model does not lose generality, as long as it is utilised with measured serum samples. Relative muscle and offal concentrations were estimated using tissue:serum partition coefficients. Unlike some PBPK models (developed for other animals like rodents and humans), the tissue concentration estimates obtained are empirical and do not provide mechanistic insights; they however provide invaluable information for risk management of livestock. Further comparison to similar models is provided in the [supplementary materials](#) (Section Comparison to Other Models and [Figure S9](#)). The partition coefficients were derived from a meta-analysis of published partitioning data (metanalysis methods and data are provided in the [Supplementary Materials Table S3](#)). The differential equations were solved using the deSolve (Soetaert et al. 2010)

Table 1
Mean environmental concentrations for PFHxS and PFOS at Site A1.

Analyte	Drinking Troughs		Soil	Grass
	2016	2017–2018		
PFHxS	10.6 ng/ml (n = 10)	6.1 ng/ml (n = 10)	3.1 ng/g (n = 39)	<0.2 ng/g (n = 10) ^a
PFOS	6.7 ng/ml (n = 10)	3.0 ng/ml (n = 10)	0.8 ng/g (n = 39)	<0.5 ng/g (n = 10)

^a Two samples with measurements close to the limit of reporting (0.3 ng/g and 0.23 ng/g).

package and exposure dynamics were integrated into the model using a base code function called *approxfun* (Becker et al. 1989) to construct the constant interpolation functions for time and exposure rate values (daily intake). To prove that the mass balance is maintained, the sum of the daily amount in the central compartment AC and daily amount excreted were subtracted from the daily intake and presented in the [supplementary materials](#) (Figure S8) for one of the field study sites. A schematic of the exposure and PopTK models is also provided in the [supplementary materials](#) (Figure S10) along with sample code in the R programming language.

2.3. Population kinetics and model parameterization

Population variability is expected to occur due to physiological and behavioural differences between individual animals. Due to the small number of studies available for beef cattle, data were collated from literature for both dairy and beef cattle and data analysis was performed

(further described in the [Supplementary Materials](#)) prior to pooling of the data. Inter-individual variability was modelled using a random effects function with normal distribution (*rnorm*). The standard mean and deviation for each parameter modelled were selected from published studies and a *meta-analysis* of partition coefficients. Bootstrap analysis was conducted to check stability of population toxicokinetic parameters and value ranges. Parameter selection is described in more detail in the supplementary material. Previous publications have shown that feed and free water intake in livestock can vary by approximately 20–30% depending on life stage, body condition, breed and other factors (Ward and McKague 2007; Appuhamy et al. 2016; Guinguina et al. 2020). To account for inter-individual variability for daily dose, a value of 30% was adopted.

2.4. Field site settings

Three cattle farms in Victoria (Australia), all impacted by historical use of aqueous film forming foams (AFFF) nearby, with different exposure dynamics, were selected for this study. The selected field study sites breed and/or agist cattle for the meat market with stock typically sold at 1 to 2 years of age. It is noted that the livestock discussed here were either acquired for further research or managed by either moving to another site or by limiting access to PFAS contaminated resources such that serum levels were reduced to acceptable levels (based on FSANZ trigger points for meat in Australia). One of the major influences of PFAS bioaccumulation in livestock is the possibility to depurate by accessing unpolluted feed and water, referred to in this work as livestock rotation potential (Mikkonen et al. 2023 Feb). The selected field study sites have different livestock rotation potential as well as different levels of PFAS

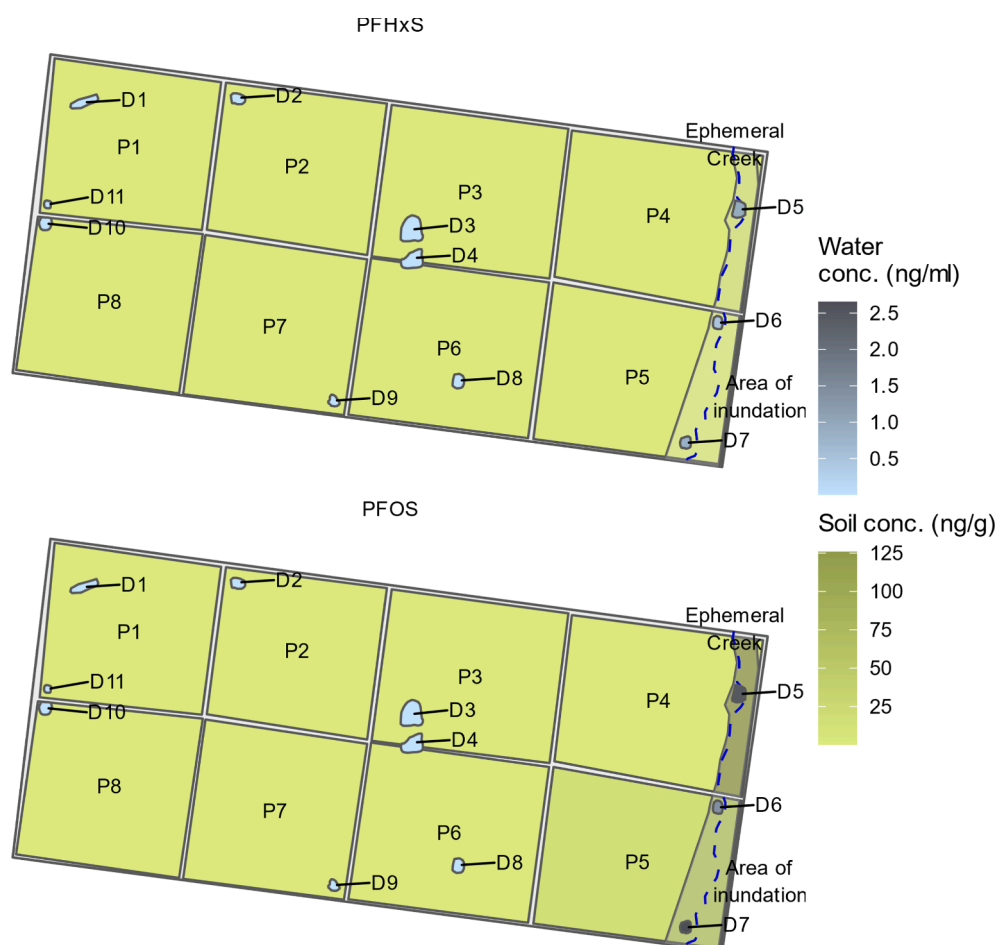


Fig. 1. Illustration of PFHxS (top) and PFOS (bottom) distribution at site A2. P1–P8 indicate paddock numbers and D1–D11 indicate surface water dams onsite.

Table 2

Mean environmental concentrations for PFHxS and PFOS at Site B1.

Analyte	Drinking water	Soil	Grass
PFHxS	2.7 ng/ml (n = 32)	17 ng/g (n = 52)	1.9 ng/g (n = 29)
PFOS	4.2 ng/ml (n = 35)	166 ng/g (n = 52)	2.1 ng/g (n = 29)

Table 3

Summary of exposure assumptions and biomonitoring data for field study sites.

Site	Exposure Assumptions	Biomonitoring
A1	Same exposure concentrations across all paddocks. Exposure ceased when animals were taken offsite.	3 rounds of serum samples following removal from site (19-, 63- and 205-days) and terminal tissue samples. Mixture of sexes and ages (3 male; 13 female), cattle were not lactating.
A2	Exposure limited by rotation time for two contaminated paddocks, low environmental concentrations on other paddocks.	1 round of serum samples collected (in Autumn 2017) approximately two months after exposure to contaminated paddock. All cattle were female and lactating (n = 10).
B1	Year-round exposure to single paddock with contaminated water, soil and pasture grass. Cattle brought onto site at 9 months of age.	4 rounds of serum samples and one round of terminal tissue sampling from 6 males.

contamination. Site A1 captures water in a dam that feeds drinking troughs across the farm, which means livestock are exposed to the same drinking water source on every paddock, while Site A2 has multiple drinking water sources, and the contamination is confined to two paddocks at one end of the site. Environmental PFAS concentrations for water, soil and pasture for both sites have been previously published (Drew et al., 2021; Mikkonen et al., 2023). At Site A1, concentrations in water, soil and pasture were similar across paddocks and are presented in Table 1. Drinking trough concentrations are presented for 2016 and later years separately due to management measures taken to limit PFAS migration from the source site.

Site A2 has an ephemeral creek and inundation zone, which connect dams across the two paddocks and acts as a mobilisation pathway for PFAS, noting that in the years under consideration no significant inundation periods were reported; PFAS concentrations were nevertheless highest in this area, indicating some inundation may have occurred in the past. Pasture grass samples were all below the limit of reporting (<LOR) from paddocks; Nevertheless, they showed elevated concentrations (ranging from 3 to 12 ng PFOS/g and 2–18 ng PFHxS/g) within the inundation zone. Paired soil and grass concentrations from the inundation zone were used to develop transfer factors (TFs) to estimate PFOS and PFHxS transfer to pasture grasses in paddocks at both sites; this was done to avoid reliance on detection limits for pasture samples. For soil samples reported as < LOR (PFHxS only LOR of 0.5 ng/g), the middle-bound method was used, replacing censored results with a value equal to LOR/2. TFs were estimated by dividing the grass concentration by the relevant soil sample concentration; TF derivation is further detailed in the Supplementary Section Soil to plant transfer factors for pasture grasses and Table S1. To maintain site anonymity, an approximate illustration of Site A2 (generated using ggplot and spatial data packages in R) is provided in Fig. 1 to show PFHxS and PFOS distribution at the site.

Site B1 also receives overland runoff from a PFAS impacted site; however, drainage channels and inundation zones distribute PFAS impacted water throughout the site, resulting in higher soil and pasture concentrations (relative to A1 and A2). Site B1 consists of a single larger paddock without structured rotation. As such, the soil, water and pasture PFAS concentrations were averaged, and are summarized in Table 2 to estimate exposure over the life cycle.

Exposures were estimated for cattle at each site, based on information available on livestock rotation and environmental PFAS

concentrations. Modelled serum and tissue concentrations were then compared to biomonitoring data from each site. Sites A1 and B1 had multiple rounds of biomonitoring data, while A2 only had one round of monitoring. Table 3 provides a summary of exposure assumptions and biomonitoring data available for each site.

Serum and tissue PFAS concentrations were previously published for each site (Mikkonen et al. 2022)(Drew et al. 2021 Nov 3). Temperature data were downloaded from the Australian Bureau of Meteorology website for the nearest monitoring station for the relevant time periods.

3. Results and discussion

3.1. Exposure dynamics

Multiple regression analysis to estimate DWI variability with growth and daily temperature fluctuations (48 samples) is described by Eqs. (9) with four rules (with an average correlation coefficient of 0.97 on the training data).

$$\text{If BW (kg)} \leq 454 \text{ and Temp (}^{\circ}\text{C)} \leq 21.1 \text{ then DWI} = -3.69 + 0.074 \text{ BW (kg)} + 0.68 \text{ Temp (}^{\circ}\text{C)} \quad (9a)$$

$$\text{If BW (kg)} > 454 \text{ and Temp (}^{\circ}\text{C)} \leq 21.1 \text{ then DWI} = 3.83 + 0.89 \text{ Temp (}^{\circ}\text{C)} + 0.0341 \text{ BW (kg)} \quad (9b)$$

$$\text{If BW (kg)} \leq 364 \text{ and Temp (}^{\circ}\text{C)} > 21.1 \text{ then DWI} = -69.11 + 2.76 \text{ Temp (}^{\circ}\text{C)} + 0.1159 \text{ BW (kg)} \quad (9c)$$

$$\text{If BW (kg)} > 364 \text{ and Temp (}^{\circ}\text{C)} > 21.1 \text{ then DWI} = -51.77 + 4.01 \text{ Temp (}^{\circ}\text{C)} \quad (9d)$$

The daily water intakes for field study sites were estimated using Eq. (9).

3.2. Model parameterisation and evaluation

For PFOS, there were no significant differences between partition coefficients for dairy and beef cattle for liver ($P = 0.17$) and muscle ($P = 0.44$) partitioning. However, there was a significant difference between kidney partition coefficients ($P = 0.03$). For PFHxS, there were significant differences between partition coefficients for dairy and beef cattle for kidney ($P = 0.005$) and muscle ($P = 0.012$) partitioning but not for liver partitioning ($P = 0.89$). Trend analysis also did not show significant relationships for PFOS; however, for PFHxS, a negative correlation was observed between the elapsed time and partition coefficients for kidney for both dairy and beef cattle (refer to supplementary materials Figures S2 and S3). This may be linked to renal excretion and reabsorption kinetics. Previous works have hypothesized that reabsorption of excreted PFAS in the kidneys may be saturable (Worley and Fisher 2015) which could mean that reabsorption is more efficient when PFHxS concentrations in serum (and urine) are lower resulting in a negative correlation between tissue and serum concentration ratios.

The statistical analysis provided in the Supplementary Materials suggests that liver and muscle tissue partitioning is likely to be similar for dairy and beef cattle for PFOS and that partitioning is not dose (environmentally relevant doses) or time dependent. For PFHxS, tissue partitioning may differ to some extent between beef and dairy cattle and partitioning to kidney may be influenced by time (with higher partitioning observed at time points closer to dosing). The meta-analysis and regression, conducted with pooled tissue partitioning data, showed that inter-study variability is significantly higher than intra-study variability with i^2 (residual heterogeneity / unaccounted variability) values typically > 90% (refer to supplementary materials Figures S4–S6) and that the sources of variability are yet to be determined. Given the significant inter-study variability, population variability was estimated using intra-study variability from the study with the largest weighted effect. Metanalysis results are summarized in Table S5 in the Supplementary

Table 4
Summary of toxicokinetic model parameters.

Compound	Parameter	Units	Mean	SD
PFHxS	PL	unitless	0.22	0.04
	PM	unitless	0.08	0.01
	DT50	days	9.3	1.31
	Vd	L/kg	0.14	0.01
PFOS	PL	unitless	1.44	0.13
	PM	unitless	0.08	0.01
	DT50 (non-lactating)	days	74.1 ^a	13.4
	Vd	L/kg	0.085	0.01
	Pmilk	unitless	0.015	0.004

^a DT50 of 56 days (Van Asselt et al. 2013) used for lactating cattle. SD is the standard deviation; PL, PM and Pmilk are the partition coefficients tissue:serum for liver, muscle and milk respectively; Vd is the volume of distribution; DT50 is the serum half-life.

Materials. Model parameters are summarized in Table 4. Given that liver partitioning is higher than that for kidney, it was chosen as the modelled endpoint for offal. Elimination half-life data were not pooled for beef and dairy cattle as lactation has been shown to increase elimination for PFOS but not PFHxS (Mogensen et al. 2015). As such, half-lives were selected as outlined in the Supplementary Table S8 to consider both lactating and non-lactating cattle for PFOS.

To ensure the model was fit-for-purpose, the parameterised population toxicokinetic (PopTK) model results were compared to reported concentrations from a dosing study which exposed cattle to PFAS contaminated feed for 28 days, followed by a 21-day depuration period

(Kowalczyk et al. 2013). The resulting estimates and measured concentrations for serum, liver and muscle are shown in Fig. 2.

3.3. Field study sites

Region-specific intake estimates for water, pasture and soil are provided in Fig. 3, which incorporated local weather data and pasture moisture content. The FWI estimates show a seasonal pattern with highest intakes in summer and autumn when temperatures are higher and pasture moisture is low. Pasture and soil intakes show less seasonal variability, though they increase with growth.

Daily intakes and site-specific environmental concentrations were used to estimate daily doses of PFOS and PFHxS over the timeframe (0–800 days), at the end of which most cattle go to market. Mean site-specific soil-to-plant TFs were estimated to be 0.54 and 0.03 for PFHxS and PFOS respectively, which are similar to those published previously for a range of forage crops and grasses (Stahl et al. 2009; Yoo et al. 2011; Wen et al. 2016; Wang et al. 2020). There is a notable difference in peak concentrations between the first and second summer peaks, which relates to growth; cattle weigh between 60 and 150 kg in the first summer and between 420 and 520 kg in the second summer period. Serum and tissue concentrations were then estimated using the PopTK model. For cattle born on-site, maternal transfer of PFOS and PFHxS was estimated based on mean reported (human) maternal-infant ratios of 40% (6 studies) and 70% (9 studies) for PFOS and PFHxS respectively (ATSDR - Toxicological Profile 2020 May 6), and the mean measured serum concentrations for mature cows from the year cattle were born. Fig. 4 provides a summary of estimated PFHxS and PFOS

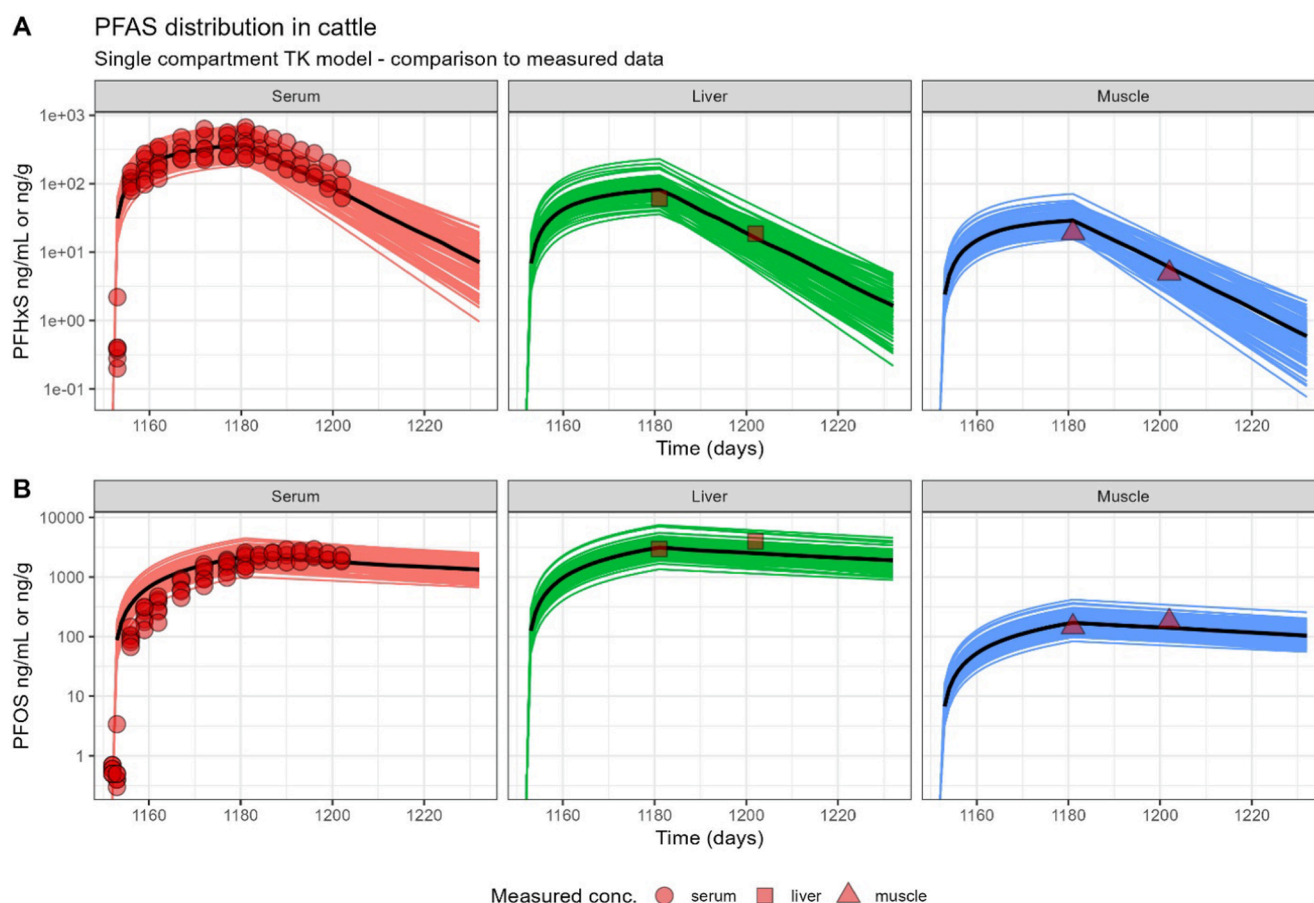


Fig. 2. Comparison of modelled and measured data for cattle (Kowalczyk et al. 2013) using the PopTK model for A) PFHxS and B) PFOS. The time represents the age of the animals (days); the experimental timeframe was 49 days. The median modelled estimate is presented as a black line and the coloured lines represent 100 individual stochastic simulations and the measured concentrations for each compartment are overlaid as red shapes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

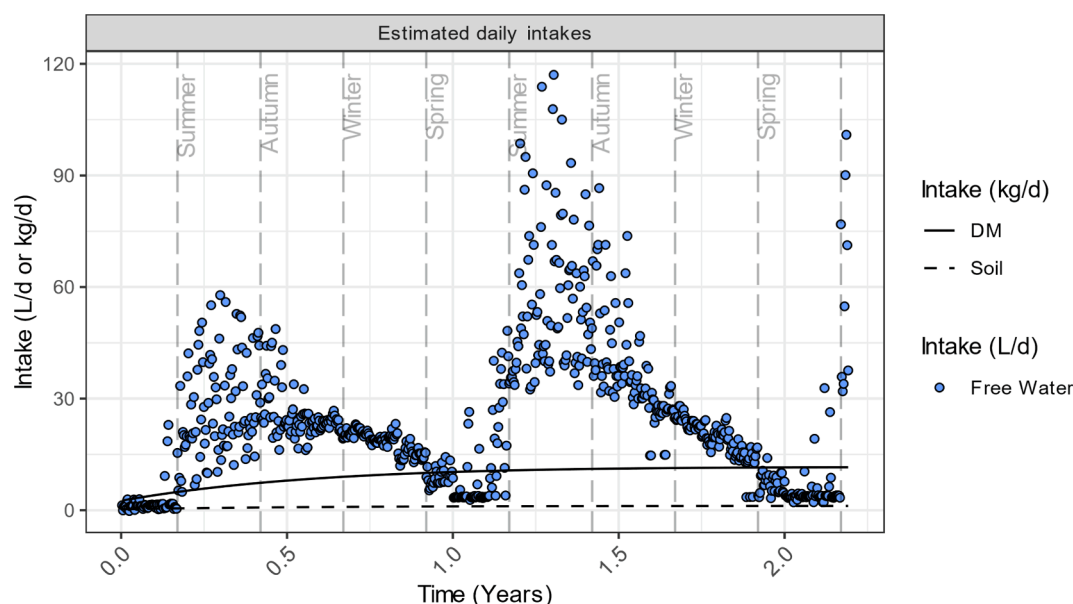


Fig. 3. Estimated daily intakes of water, pasture (as dry matter or DM) and soil for cattle.

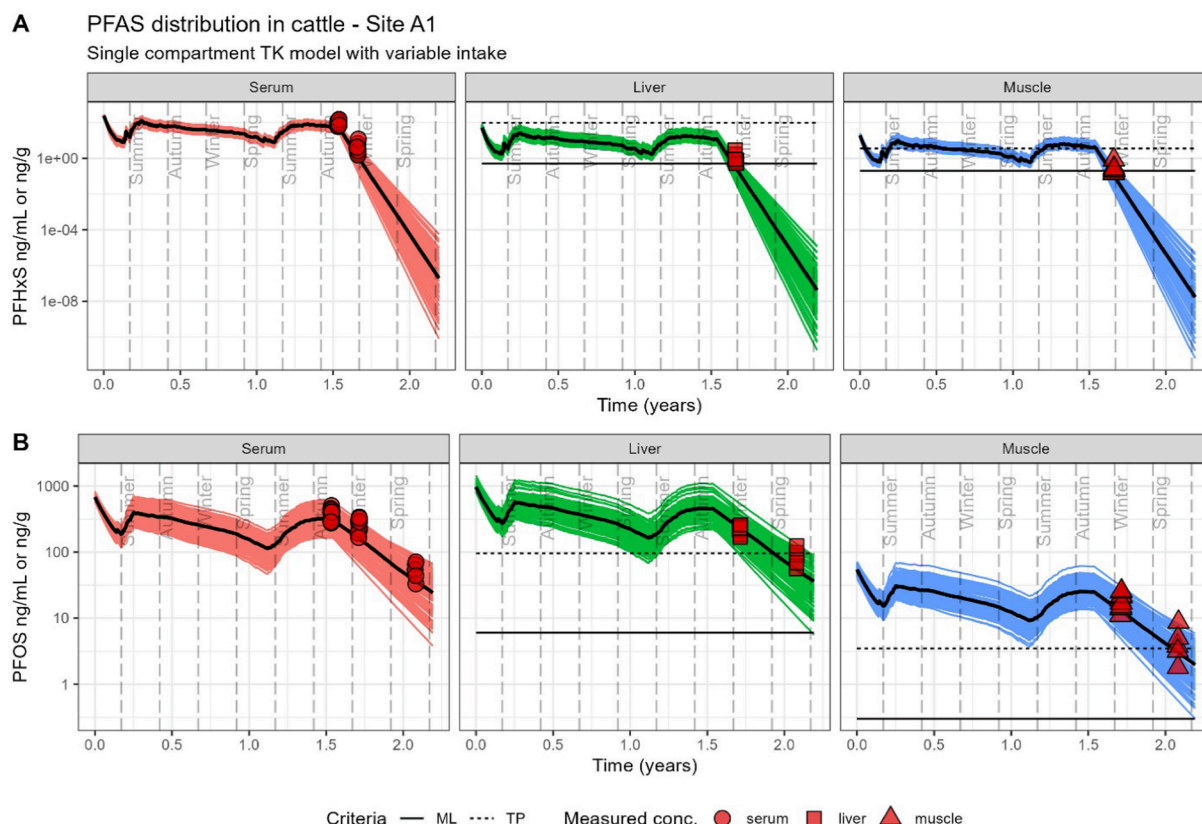


Fig. 4. Estimated serum, liver and muscle concentrations for cattle exposed to PFAS at Site A1 for 1.5 years followed by a depuration period (approximately 200 days) for A) PFHxS; B) PFOS. Maternal transfer is based on serum concentrations (average) from mature cattle collected in 2016 prior to runoff mitigation works on the source site (PFOS = 1800 ng/mL and PFHxS = 400 ng/mL). The median modelled estimate is presented as a solid black line within the coloured lines that represent 100 individual stochastic simulations and the measured concentrations are overlaid for each as solid red shapes. Guideline comparisons are provided for the EC MLs (solid straight line bovine muscle meat: 0.3 ng PFOS/g and 0.2 ng PFHxS/g; bovine offal: 6 ng PFOS/g and 0.5 ng PFHxS/g) and FSANZ TPs (dotted straight line; 3.5 ng PFOS and PFHxS/g meat and 96 ng PFOS and PFHxS/g offal). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

concentrations over time for cattle at Site A1, and a comparison to maximum levels (MLs) for offal and muscle meat from the European Commission (EC) (EC 2022) and Food Standards Australia New Zealand

(FSANZ) (FSANZ 2021). It is noted that these guidance values are not promulgated guidelines (at the time of writing), and the FSANZ trigger points (TPs) are intended for use during contaminated site investigations

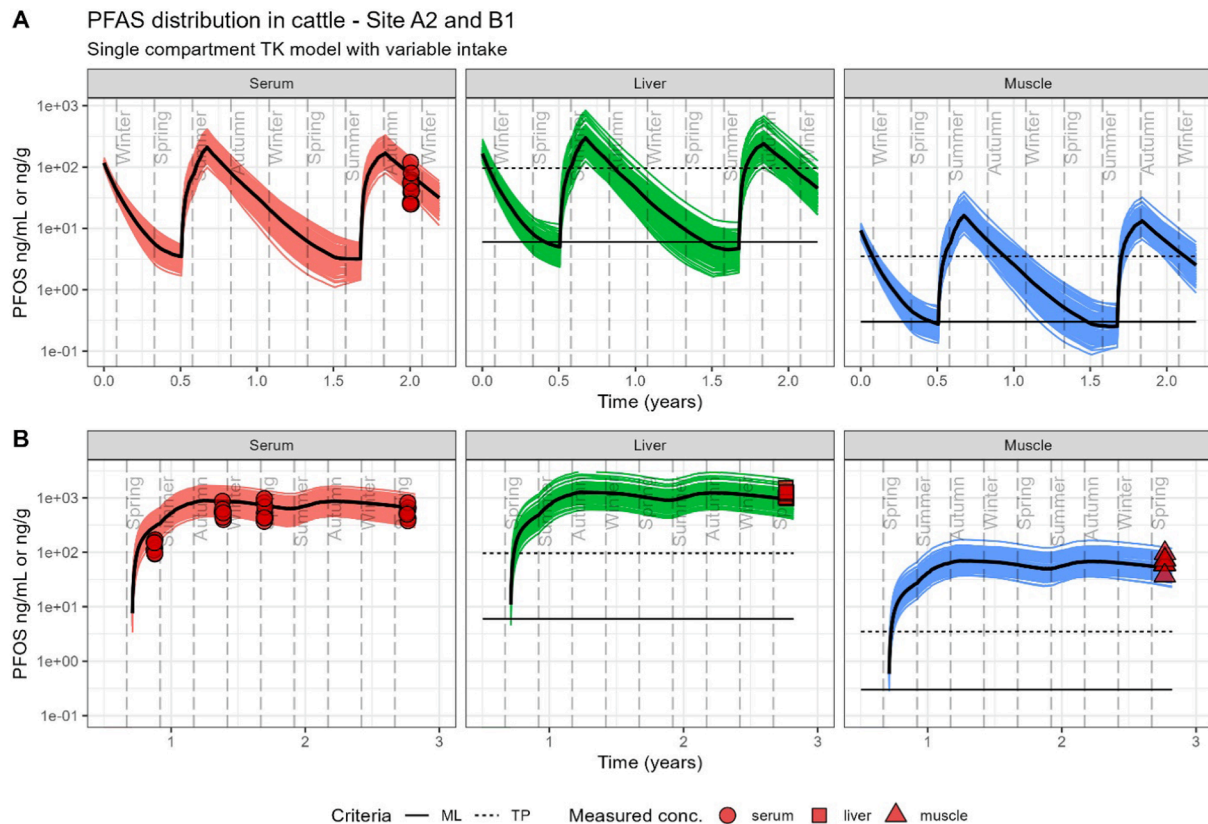


Fig. 5. Estimated and measured serum and tissue concentrations for cattle exposed to PFAS at A) Site A2 and B) Site B1 for PFOS. Maternal transfer for cattle born at Site A2 are based on modelled serum concentrations for exposure to paddocks P4 and P5 (PFOS = 125 ng/ml). Cattle were brought onto Site B1 at 9 months of age. The median modelled estimate is presented as a solid black line within the coloured lines that represent 100 individual stochastic simulations and the measured concentrations overlaid as solid red shapes. Guideline comparisons are provided for the EC MLs (solid straight line) and FSANZ TPs (dotted straight line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for potentially impacted farms, to identify what local risk management options might be considered appropriate.

The estimated serum and tissue concentrations at Site A1 show a similar seasonal trend as water intake, which would suggest that the contribution to daily dose from water is significant. The trend is more pronounced for PFHxS, due to its shorter half-life. The comparison to EC MLs shows that the median muscle and liver concentrations exceed both the local (FSANZ) and European Commission MLs for PFOS at all times. Owing to a shorter half-life (Kowalczyk et al. 2013; Drew et al. 2021 Nov 3) and lower tissue partitioning (compared to PFOS), liver and muscle PFHxS concentrations show greater fluctuations and, by the end of spring, have fallen to concentrations close to the EC MLs. However, it is noted that both the EC and FSANZ guidance values also limit the sum of the individual PFAS, so based on PFOS concentrations alone these limits are exceeded.

At Site A2, PFAS exposures are intermittent, and as such, the estimated serum and tissue concentrations peak when cattle are rotated through the contaminated paddocks and then decline until cattle are rotated back onto the contaminated paddocks again. In this case, cattle were rotated through the contaminated paddocks in the summer months, when water intakes and PFAS exposures were highest. Exposures at Site B1 are less impacted by seasonal variability due to elevated PFAS concentrations in soil and pasture. Considering PFOS as the risk driver, modelled serum and tissue concentrations for PFOS only are provided in Fig. 5 for Sites A2 and B1. Comparison to guidance values shows that offal and muscle PFOS concentrations at A2 are likely to exceed FSANZ TPs for short periods of time (3–4 months) following rotation at P4 or P5, however, are likely to exceed EC MLs for 10–12 months following exposure. At Site B1, muscle and offal PFOS

concentrations were estimated to exceed guidelines within a week from start of exposure and remain elevated throughout their lifecycle at Site B1.

3.4. Interim exposure management measures

Given that the modelled (and measured) concentrations exceed maximum levels for each site, interim management measures (cattle rotation, feed/water supplementation) are required, while longer term solutions like remediation options are assessed. Considering that water is the primary driver for PFAS exposure at A1, the only interim management measures that would significantly reduce exposure would be to replace the contaminated water source (which feeds all of the drinking troughs onsite) or remove cattle from the site altogether. At A2, exposure to PFAS is intermittent and could be further reduced by avoiding cattle rotation through the contaminated paddocks (P4 and P5), or by timing the rotation to occur in the winter or spring months (when free water intake is lowest). Modelling can also be used to design an optimal rotation schedule that ensures compliance with MLs for the relevant market.

Fig. 6 shows the expected reduction in PFOS serum and tissue concentrations for the interim exposure management measures discussed for each site. The results show that the substitution of water sources on Site A1 could reduce tissue concentrations down to below FSANZ TPs, but would require further elimination controls, such as soil remediation, for the herd to meet the EC MLs within normal farming timeframes. For Site A2, the rotation of cattle through contaminated paddocks in winter and earlier in their life cycle (without repeat rotation) was estimated to result in median tissue concentrations below FSANZ and EC guidance

A PFOS distribution in cattle - management/mitigation scenarios

Single compartment TK model with variable intake

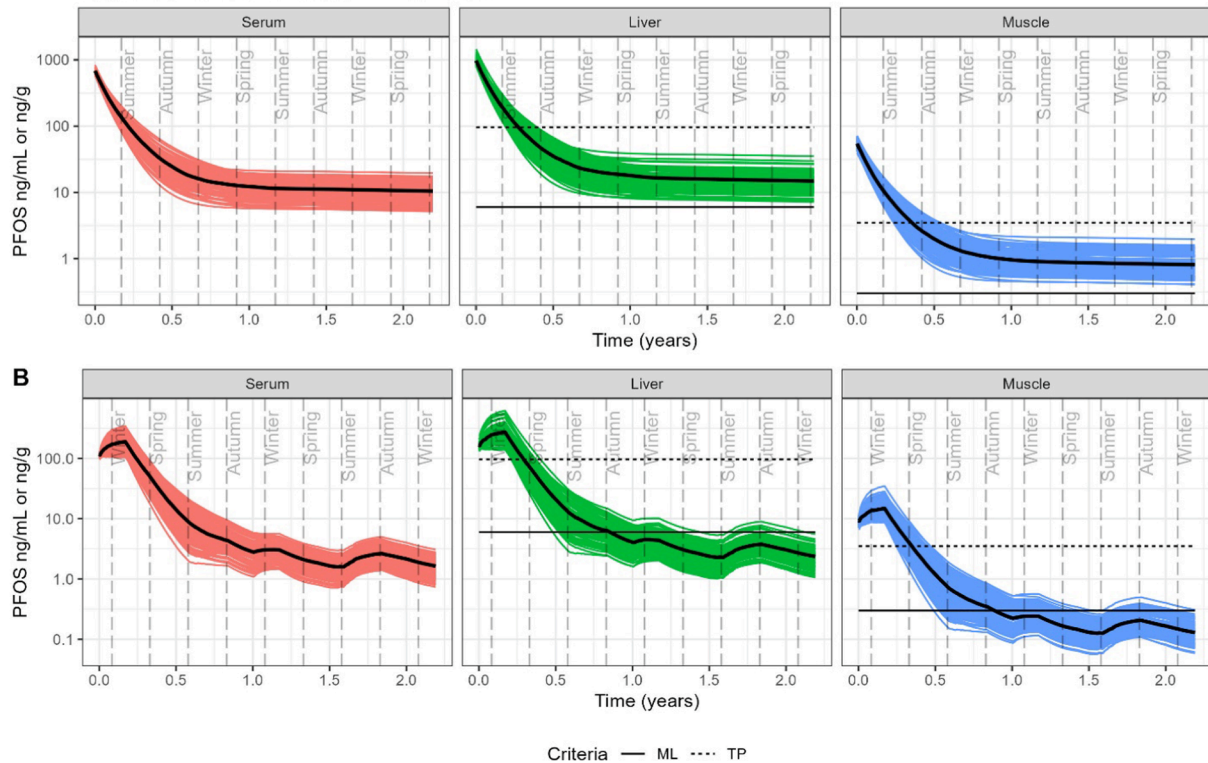


Fig. 6. Modelled outcomes of the interim exposure management measures as the estimated serum (central), liver and muscle PFOS concentrations for cattle born onsite at A) Site A1 and B) Site A2. Maternal transfer was based on estimated maternal serum concentrations (A1 PFOS = 1800 ng/ml; A2 PFOS = 125 ng/ml). The median modelled estimate is presented as a solid black line within the coloured lines that represent 100 individual stochastic simulations. Guideline comparisons are provided for the EC MLs (solid straight line) and FSANZ TPs (dotted straight line).

PFAS distribution in cattle - Sweden

Single compartment TK model with variable intake

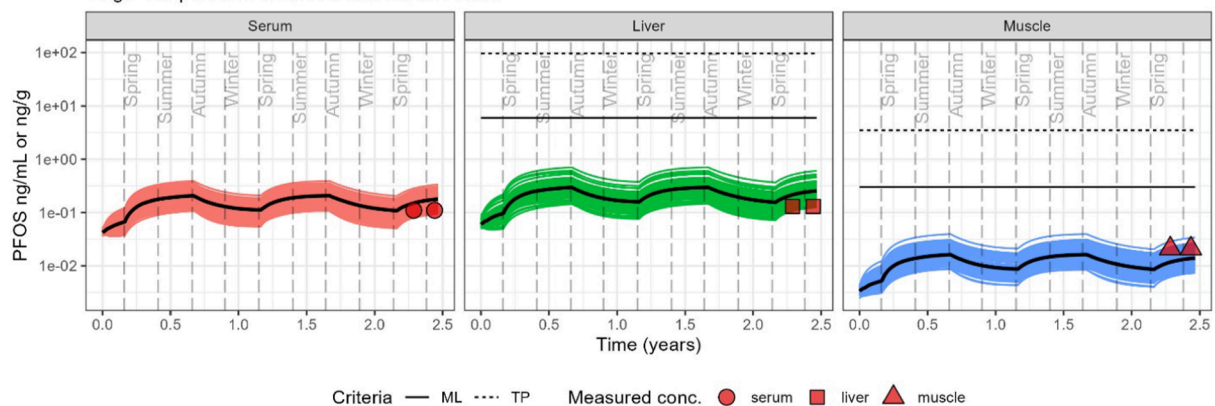


Fig. 7. Estimated serum (central), liver and muscle PFOS concentrations for dairy cattle in Sweden (based on published data (Vestergren et al. 2013)) exposed to low background concentrations. Maternal transfer based on estimated maternal serum concentrations (PFOS = 0.11 ng/ml). The median modelled estimate is presented as a solid line within the coloured lines that represent 100 individual stochastic simulations. Guideline comparisons are provided for the EC MLs (solid straight line) and FSANZ TPs (dotted straight line). Measured blood, liver and muscle concentrations are shown for the reported averages of two collection points.

values, suggesting that cattle exposure could be successfully managed on this site. Maternal serum levels also have a significant impact on offspring tissue concentrations and the timeframe for market-ready cattle. For A1 and A2 water supplementation and rotation controls would be expected to reduce maternal PFAS concentrations and transfer. For Site B1, the only viable short-term management measure was to remove cattle from the site, considering that the PFAS contributions were significant from all three sources: water, pasture and soil.

3.5. Model applicability and considerations

Additional considerations may apply for different cattle rearing conditions, such as those experienced in feedlots or winter housing, where the diet may be more heavily supplemented (with hay, grains and silage). The PFAS content may be variable if roughage and grains are sought from multiple sources. Feed and free water intake, as well as exposure to soil, may also vary during the winter housing period in

colder regions, which may impact exposure to contaminants. As such, field programs should take into consideration how exposures may vary through the year; for example, several monitoring rounds may be preferable to account for possible seasonal variability while water quality or quantity is changes.

Exposure equations can also be modified to suit conditions and consumption patterns for different regions and cattle breeds as necessary. As an example, Fig. 7 provides estimated serum, liver and muscle concentrations for dairy cattle from Sweden, for a site described by Vestergren et al. (Vestergren et al. 2013) that had low background PFOS concentrations. Adapted exposure equations from Cardot et al. (Cardot et al. 2008) for dairy cattle were used. Fig. 7 assumes that the cows spend 6 months in indoor housing and 6 months grazing on paddocks, which shows a possible seasonal trend associated with the difference in PFAS concentrations in silage and pasture grasses as well as incidental ingestion of soil during grazing. The good agreement of the simulated and measured concentrations shows that PopTK may be applicable also at these lower concentrations, much lower than the model was parametrised with originally.

4. Conclusions

This study developed and validated a simple approach to estimate daily cattle exposure to PFOS and PFHxS from the farm environment, incorporating variables over time such as cattle growth, seasonal behaviours and changes in contamination levels between farm areas. The exposure data was then fed into the population toxicokinetic (PopTK) model to estimate the resulting blood and tissue concentrations in cattle over time. The ability to incorporate farm-specific daily intake considerations allowed for a greater degree of flexibility in estimating the PFAS accumulation potential (for cattle) and allowed for consideration of interim exposure management measures over the life of the animals to reduce PFAS concentrations to the extent practicable. A final consideration for the use of these models is that biomonitoring data (like serum) should be collected to check estimated serum or tissue concentrations, validate management options and verify suitability for consumption.

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CRedit authorship contribution statement

Antti T. Mikkonen: Conceptualization, Methodology, Formal analysis, Visualization, Writing – original draft. **Jennifer Martin:** Investigation, Resources, Writing – original draft. **Richard N. Upton:** Methodology, Validation, Supervision. **Jan-Louis Moenning:** Validation, Writing – review & editing. **Jorge Numata:** Validation, Writing – review & editing. **Mark Patrick Taylor:** Writing – review & editing. **Michael S. Roberts:** Writing – review & editing, Supervision. **Lorraine Mackenzie:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Antti Mikkonen reports a relationship with Nation Partners Pty Ltd that includes: employment. Jennifer Martin reports a relationship with Arcadis that includes: employment.].

Data availability

All data is referenced and data compilations are explained in the [supplementary materials](#)

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2023.108218>.

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